

**Contrasting patterns of juvenile chinook salmon
(*Oncorhynchus tshawytschaw*) growth, diet, and prey densities
in off-channel and main stem habitats on the Sacramento
River.**

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Abstract

Few studies have quantified juvenile salmon growth in or between different habitats or evaluated the specific mechanisms by which salmon growth and survival might be enhanced. We utilized otolith microstructure to estimate daily relative growth rates among main channel areas, off-channel ponds, and non-natal seasonal tributaries of the Sacramento River. To examine possible mechanisms leading to growth differences, prey availability, prey preference, and stomach fullness were estimated at each site. We observed wider daily increment widths, higher prey densities, and warmer temperatures in off-channel ponds and non-natal seasonal tributaries in both 2001 and 2002.

Chironomidae larve, pupae, and adult diptera were the preferred prey in all habitats. No difference in stomach fullness was found between habitats. In 2001, all habitats had higher temperatures, wider daily increment widths, higher prey densities, and higher stomach fullness than in 2002. Our findings suggest the warmer temperatures and abundant prey in off-channel habitats leads to higher growths rates, which in turn can lead to higher juvenile survival rates.

Introduction

Many wild salmonid stocks are exhibiting precipitous declines around the world (Netboy 1980, Moyle and Cech 2000). This is particularly evident in California where subspecies, and/or populations of three anadromous salmonids [chinook salmon (*Oncorhynchus tshawytscha*), coho salmon (*O. kisutch*) and steelhead trout (*O. mykiss*)] currently have state or federal protection (Yoshiyama et al. 2000). Salmonid declines can be quite dramatic; Yoshiyama et al. (2000) have documented a 75% decrease in overall chinook salmon abundance in California's Central Valley since 1950. Models by Karieva et al. (2000) on Columbia River chinook salmon suggest that first year and estuarine survival are the key factors influencing a cohort's success. While regional differences exist between river systems, young of the year survival rates are undoubtedly an important aspect of stock recruitment. Unfortunately, our current understanding of how large river processes influence salmonid rearing success is incomplete (Vogel and Marine 1991). Swales et al. (1986) suggest a need for more information on habitat requirements of juvenile salmonids in the hopes of identifying factors limiting their abundance during the freshwater phase.

In addition to rearing in the main channel of rivers, juvenile salmon have been observed rearing in multiple habitats including; floodplains (e.g. Sommer et al. 2001), off-channel ponds (e.g. Peterson 1982), natal tributaries (e.g. Johnson et al. 1992), and non-natal tributaries (e.g. Murray and Rosenau 1989). It is speculated that salmon move into these off-channel habitats for their improved rearing conditions. Refuge from high flows and high sediment loads may improve growth rates (Crouse et al. 1981) and decrease mortality (Erman 1988). Upon inundation the increase in off-channel rearing area may reduce competition and lower the chance of encountering a predator (Sommer et al. 2001). Greater prey densities relative to the main channel may also improve

feeding rates and result in faster growth (Swales and Levings 1989, Sommer et al. 2001). Unfortunately, the effects of off-channel habitats on juvenile salmon have rarely been quantified (Simenstad and Cordell 2000).

Previous work on the growth rates of juvenile salmon in the wild has generally followed two distinct methodologies: change-in-length and mark-recapture. Change-in-length methods estimate growth rates by comparing the mean or median length of a population at two points in time. Populations within a habitat are selected from length-frequency distributions. The method assumes no size-selective predation or sampling (Swales et al. 1986). Mark-recapture methods avoid assumptions of sampling error but can be costly when applied to large populations and frequently yield low return rates (e.g. Sommer et al. 2001).

Analyzing otolith microstructure is a promising alternative for examining growth rates in the wild (Miller and Simenstad 1997). Daily growth increment widths on sagittae otoliths are positively correlated with salmon growth rates (Neilson and Geen 1982, Neilson et al. 1985). One early concern with the method was whether increment widths might be more influenced by temperature than somatic growth (Neilson and Geen 1982, Bradford and Geen 1992), but Gauldie (1991) demonstrates that changes in increment width do not correspond to predicted values based on temperature effects alone. The evidence suggests otolith microstructure provides a conservative estimate of somatic growth and is a useful tool for looking at short-term relative growth differences (Neilson et al. 1985, Gauldie 1991).

In the present study we use otolith daily growth increments as a relative measure of somatic growth in fall-run juvenile chinook salmon. We hypothesize that off-channel habitats provide improved rearing conditions for juvenile chinook salmon relative to the main channel. Specifically, we believe the higher temperature, clarity, and shallow depth

of off-channel waters will lead to higher prey densities and higher salmon growth than in the main channel. We test this hypothesis by comparing juvenile salmon growth, diet, and stomach fullness, along with prey abundance, temperature, and turbidity in the main channel, off-channel ponds, and non-natal seasonal tributaries of the Sacramento River.

Methods

Study Area

The Sacramento River is the largest river system in California and perhaps one of the most disrupted in the world (Yoshiyama et al. 2000). The approximately 70,000 km² watershed is heavily altered by dams and diversions primarily for agriculture and urban development (Reisner 1986, May and Brown 2002). Historic discharge near Red Bluff, CA ranges from summer lows of 28 m³/s to winter highs of 11,900 m³/s (CDWR 1994a). The river originates near Mt. Shasta and is fed primarily by snowmelt and precipitation runoff. Over our two year sampling period discharge near the study area ranged from 170 m³/s to 2478 m³/s (Hamilton City station, CDWR). All sampling took place between the towns of Los Molinos and Ord Bend (river miles 224 and 168 respectively, Figure 1).

The Sacramento River supports four recognized runs of chinook salmon: winter, fall, late-fall, and spring runs. The fall-run is typified by a stream-type life history (Healey 1991) and is currently the numerical dominant of the four runs (Yoshiyama et al. 2000). Fall-run adult migration peaks during September and October and spawning occurs soon after adults reach their natal stream (Vogel and Marine 1991). After emerging in winter and early spring, the fall-run fry typically rear in main stem rivers or the bay-delta estuary before heading to the ocean (Kjelson et al. 1982).

Fish sampling

We collected fall-run juvenile salmon from 3 different habitat types: main channel, off-channel ponds, and non-natal seasonal tributaries. We sampled during March-April in 2001 and February-March in 2002 using 10-m and 15-m beach seines (4.75-mm mesh). Sampling occurred between 0730h to 1100h to minimize differential digestion of prey

items. In both years we visited each site three times and collected ten fish each visit, for a total of 30 fish site⁻¹ year⁻¹.

In 2001 we sampled two main channel sites and one off-channel pond site. The two main channel sites (MC 1 and MC 2) were along gravel bars on the inside of a meander bend (Figure 1). Gravel bars were bordered by a mixed riparian forest dominated by cottonwoods, sycamores, and willows. Our off-channel pool site (OP 1) contained approximately 3100 m³ of water and was continuously connected to the main channel at the downstream end (Figure 1). During the first week of sampling high flows inundated off-channel pond 1 at the upstream end for 3 days. Willows were the dominant vegetation around off-channel pond 1.

In 2002 we sampled three main channel sites, three off-channel pond sites, and two non-natal seasonal tributary sites (hereafter referred to as seasonal tributaries). Main channel sites (MC 3, MC 4, and MC 5) occurred on gravel bars 100 meters upstream from a off-channel pond (Figure 1). Willows and cottonwoods were dominant around all main channel sites. The size of off-channel pond sites ranged from approximately 2800 m³ (OP 2) to 5500 m³ (OP 3). OP 1 and OP 2 lost connection to the main channel during the last week of sampling when discharge in the main channel dropped below 225 m³/s. Our third site (OP 3) maintained connection to the main channel throughout the study period. Willows were the dominant vegetation around all off-channel ponds.

We describe the seasonal tributaries as non-natal, based on evidence of salmon presence but not rearing in a study by Maslin et al (1997). Seasonal tributaries were connected to the main channel by short periods of surface flow (typically December through May) and were typified by quick responses to precipitation (flashy). The Toomes Creek site (ST1) is situated in a great valley mixed riparian forest habitat dominated by cottonwoods, sycamores and willows, while the Mud Creek site (ST2) is

situated in a valley freshwater marsh with willows and grasses as the predominant vegetation (Figure 1).

Otoliths

The weight and standard length for each fish was taken prior to otolith removal. The sagittae otoliths were removed and washed in 95% ethanol. Once dried, the right side otoliths were mounted on microscope slides in Crystalbond™ with the sulcus acousticus facing down. On 6 occasions (N= 330) the calcium carbonate structure of the otolith was in the vauterite form (rather than the more common aragonite) which obscures daily increments and the left sagittae otolith was used.

We polished otoliths on both sides (see Secor et al. 1992) down to the primordia using 600 wet grit sand paper followed by alumina micropolish (0.05µm grit, Buehler ltd.). Polishing continued until daily increments were clearly visible using light microscopy. Each mounted otolith was assigned a random number to prevent bias during later analysis.

We photographed otoliths at 400X using a Pixera Penguin digital camera mounted to an Olympus BX-51 compound microscope. The ten most recent daily increment widths were measured using Metamorph® imaging analysis software and an average daily increment width (here after referred to as increment width) was calculated for each fish. All measurements were made at a 45° angle to the longitudinal axis at the posterior end (Figure 2).

Diet

After otolith removal the stomach of each fish was dissected, patted dry, and weighed. After removing stomach contents the empty stomach was re-weighed to determine stomach content mass. Stomach contents were placed in 95% ethanol for preservation. Prey item identification followed Merritt and Cummins (1984) and Borror et al. (1989). Aquatic insect larvae and pupae were identified to family, while aquatic adult insects, terrestrial insects, and crustaceans were identified to order. Due to the difficulty of distinguishing among aquatic and terrestrial dipteran adults, all were classified as winged diptera.

Individuals of the same taxa were grouped in a petri dish and the percent volume of total contents for each taxa was visually estimated. An Index of Relative Importance (IRI) was calculated for samples from each habitat using the frequency of occurrence, frequency by number, and percent volume for each prey according to Shreffler et al. (1992). The formula for IRI is as follows:

$$\text{IRI} = \text{freq. of occurrence} (\text{freq. by numbers} * \text{percent volume}) \quad (1)$$

Due to small stomach size, we used percent volume rather than the standard measure of percent biomass (Shreffler et al. 1992). Although possible error is introduced when visually estimating volume, we feel this method may reduce our bias because of problems associated with sclerotized structures in the gut contents which are less prone to digestion and are heavier than softer tissues (Hyslop 1980).

An index representing stomach fullness (FI) was calculated by dividing the wet weight of stomach contents (sc) by the wet weight of the juvenile salmon (Miller and Simenstad 1997). The formula for FI is as follows:

$$FI = WW_{sc} / WW_{salmon} * 100 \quad (2)$$

Prey availability

Invertebrates were collected on every sampling occasion using 13 cm diameter plankton nets with 263 μ m mesh. Ten plankton net tows were made at different points within the habitat to sample microhabitat variation including depth and substrate. Each plankton tow was 10 m in length. The ten samples were combined into a single composite sample and preserved in 90 % ethanol. Due to high numbers of organisms three samples from backwater sites in 2001 and 2002 were subsampled. After measuring the total sample volume we agitated the sample for 10 seconds and removed a subsample of known volume. Subsampling continued until 300 organisms were counted. Identification followed the same procedures as for stomach contents.

Physical Conditions

We measured discharge, turbidity and temperature at all sites. Discharge measurements in the main channel were obtained from the Hamilton City gauge station operated by the California Department of Water Resources. Discharge measurements in the tributaries were made manually according Nielson and Johnson (1983). We made discrete temperature measurements using a thermometer in both years. In 2002, temperature loggers (Onset Corporation) were placed in all habitats and temperature measurements were made every hour. Mean daily water temperature was calculated from the 24 measurements on the day of sampling. We measured turbidity using a DRT 15-CE Portable Turbidimeter.

Analysis

We performed analyses of variance (ANOVA) with a nested design (site nested within habitat) to test hypotheses about salmon increment width and stomach fullness in different habitats. We used Tukey's multiple comparison test to compare habitat means at a 0.05 significance level. Due to possible correlation between increment width and fish length, we compared salmon lengths between habitats using a nested ANOVA. If differences were present, ANCOVA was used to test for interaction between salmon length and habitat. We compared prey density using a nested ANOVA. For the comparison we limited prey taxa to those that constituted 96% IRI or greater in the salmon stomachs. We evaluated prey preference by running a Spearman's rank correlation procedure comparing stomach contents to plankton net samples, testing the null hypothesis of no correlation. We tested for differences between discrete temperature measurements and daily averages from temperature loggers using Wilcoxon's matched-pairs test. Water temperature and turbidity between habitats were compared using a nested ANOVA. We analyzed the importance of habitat, year, stomach fullness, prey density, temperature, and turbidity in predicting increment width for both years using a general linear model. Interactions between variables were not tested due to the limited degrees of freedom. All statistical analysis was done using JMP 5.0 (SAS Institute Inc.).

Results

Fish sampling

Our use of beach seines coupled with fish schooling behavior and large habitat area prevents us from quantifying salmon densities. We did however; observe higher numbers of salmon and other fish caught per seine haul in off-channel habitats. In addition to juvenile salmon we collected high numbers of native species in both off-channel ponds and non-natal tributaries, as did Maslin et al. (1997) and Moore (1997). We collected Sacramento sucker (*Catostomus occidentalis*), Sacramento pikeminnow (*Ptychochelilus grandis*), and roach (*Lvinia symmetricus*), all of which are known to use intermittent streams for spawning and rearing (Moyle 2002). No largemouth bass (*Micropterus salmoides*), bluegill (*Lepomis macrochirus*), or green sunfish (*Lepomis cyanellus*) were captured in the non-natal tributaries, suggesting that their life history is either less adapted to invading these habitats or no local sources were close by. All three introduced centrachids were present in the off-channel ponds throughout the year. Also captured in off-channel ponds were Sacramento sucker (*Catostomus occidentalis*), Sacramento pikeminnow (*Ptychochelilus grandis*), prickly sculpin (*Cottus asper*), golden shiner (*Notemigonus crysoleucas*), and inland silverside (*Menidia beryllina*).

Relative Growth Rate

In 2001 we measured a significant difference in increment width between habitat types ($F = 14.15, p < 0.0001$). Off-channel pond salmon had larger increment width than main channel salmon (Figure 3). Both habitats had similar mean salmon lengths ($F = 0.30, p = 0.74$). In 2002 increment width again differed between all habitat types ($F = 33.44, p < 0.0001$). Off-channel pond salmon had the largest increment width followed

by seasonal tributary salmon (Figure 3). Average salmon length differed between habitats in 2002 (Table 1), but the interaction with habitat type from ANCOVA results was not significant ($F = 0.05$, $p = 0.95$), suggesting the difference between habitats was independent of salmon length. Similar habitat types had different increment width between years ($F = 40.62$, $p < 0.0001$), with higher values seen in 2001 in both off-channel ponds and the main channel.

Habitat, prey density, and water temperature had significant effects on increment width in both 2001 and 2002 based on general linear models (Table 2). Turbidity and stomach contents had significant effects on increment width in 2001 but not in 2002.

Diet

Juvenile salmon preferred aquatic dipterans in all habitats (Figure 4). In 2001 off-channel pond salmon preferred chironomidae larvae, pupae, and diptera adults (totalling 69% IRI) followed by cladocera (30% IRI). Main channel salmon diets also showed a preference for chironomids and adult diptera (totalling 78% IRI) followed by aphids (15% IRI). Diets in 2002 were again dominated by chironomidae larvae and pupae, with values of 52%, 97%, and 89% for off-channel pond, main channel and seasonal tributary salmon, respectively. Copepods were also a significant prey item in off-channel pond salmon stomachs (46%). Hydropsychidae larve (Trichoptera), baetidae nymphs (Ephemeroptera), and arachnids were consumed infrequently in both years.

We observed no difference in stomach fullness in 2001 ($F = 2.93$, $p = 0.06$, Figure 5). In 2002 we rejected the null hypothesis of equal stomach fullness between habitats and sites ($F = 4.5$, $p < 0.0001$), but the effect of habitat was not significant ($p = 0.19$) and multiple comparison tests showed no difference between habitats. Differences were

found between years, with higher stomach fullness in both off-channel pond and main channel salmon in 2001 than in their respective habitats in 2002 ($F = 21.0, p < 0.0001$).

Prey Availability

Mean prey densities were higher in off-channel habitats in both years but the results were not always significant (Figure 6). Prey densities were significantly higher in the off-channel ponds than in the main channel in both 2001 ($F = 9.46, p = 0.02$) and 2002 ($F = 2.71, p = 0.05$, Tukey's). Seasonal tributaries has similar prey densities to both off-channel ponds and the main channel in 2002.

Cladocera and copepods were the dominant taxa found in both habitats followed by chironomidae larvae (Figure 7). In both years cladocera and copepod densities were over a magnitude greater in off-channel ponds than in the main channel, with a reversal in the predominant type between years (copepods in 2001 and cladocera in 2002). Between years main channel habitats had higher densities in 2001 while off-channel ponds had higher densities in 2002.

Chironomidae larvae were the second most common prey in all habitats followed by chironomidae pupae (Figure 7). In 2001 chironomidae larvae densities in off-channel ponds were similar to those in main channel habitats, while in 2002 they were three times the main channel density and one third the seasonal tributary density.

Using Spearman's rank correlation procedure we failed to reject the hypothesis of no-correlation between diet and plankton samples in any habitat (Table 1). Diets were either correlated to what was present in the water column or the variation in our samples was such that we were not able to detect differences.

Physical Measurements

Precipitation timing and high flow events on the Sacramento River differed between years. Higher precipitation (173 cm^3) occurred in 2001 and seasonal high flows were in early March ($2095 \text{ m}^3\text{s}^{-1}$). In 2002 precipitation was lower (97 cm^3) with the seasonal high flows occurring earlier in January ($2478 \text{ m}^3\text{s}^{-1}$). Seasonal tributary discharge ranged from $1.72 \text{ m}^3/\text{s}$ to $2.81 \text{ m}^3/\text{s}$ in Tributary 1 and $0.80 \text{ m}^3/\text{s}$ to $2.12 \text{ m}^3/\text{s}$ in Tributary 2 during the study period.

No differences were found between discrete temperature measurements and daily averages calculated from temperature loggers ($p > 0.56$). In 2001 mean water temperatures were higher in off-channel pond 1 than in the main channel habitats ($F = 9.31, p = 0.01$, Figure 3). In 2002 all off-channel pond and main channel water temperatures were lower than their respective 2001 values ($F = 15.12, p < 0.001$, Tukey's). Water temperatures in 2002 differed between habitats ($F = 3.13, p = 0.03$, Figure 3), with the highest temperatures found in the off channel pools. Based on multiple comparisons off-channel pond temperatures were significantly higher than main channel habitats but not different from seasonal tributaries. Turbidity values were lower in off-channel habitats in both years but the differences were not significant (2001, $F = 1.05, p = 0.41$; 2002, $F = 1.77, p = 0.16$, Table 1).

Discussion

Previous research has often inferred benefits from differential rearing habitat based on salmon presence (Simenstad and Cordell 2000). In the present study, we document higher chinook salmon growth rates in off-channel habitats of the Sacramento River. Our results using otolith daily increment width agree with previous growth studies using change-in-length (Swales et al. 1986, Murray and Rosenau 1989) and mark/recapture techniques (Sommer et al. 2001). Based on otolith microstructure we conclude that during the two-year study, juvenile chinook salmon rearing in off-channel habitats (both off-channel pools and non-natal tributaries) grew faster than salmon rearing in the main channel.

Salmon rearing in off-channel habitats experienced warmer water temperatures. Warmer temperatures can result in higher metabolic rates and faster growth when adequate food is present. In the laboratory under maximum ration conditions, Central Valley salmon growth rates increase with increasing water temperature up to 19°C (Rich 1987, Marine 1997, Cech and Myrick 1999). In our study of wild salmon, higher growth rates were associated with warmer conditions within a year as well as between years. Within years, salmon growth rates and water temperatures were consistently higher in off-channel habitats. Between years, we observed higher salmon growth rates and water temperatures in both off-channel ponds and main channel habitats in 2001. Main channel salmon in 2001 grew at a similar rate to salmon rearing in off-channel ponds in 2002. We conclude that temperature played a major role in salmon growth rate differences between habitats, and seasonal variation in water temperature could have significant impacts on salmon growth.

Prey availability may be another factor influencing salmon growth rates. Higher metabolic rates require adequate food for higher growth rates to result. We observed higher prey densities in the warmer off-channel habitats in both years, results consistent with other research on floodplains (Gladden and Smock 1990, Sommer et al. 2001) and shallow habitat interfaces (Welcomme 1979). Diptera densities were similar between habitats in 2001 and significantly higher in off-channel habitats in 2002. Zooplankton densities were higher in off-channel habitats in both years. We expected to see higher stomach fullness in habitats with greater prey densities due to higher feeding rates. Stomach fullness was higher in both habitats in 2001, but we did not see fuller stomachs between habitats with different water temperatures and prey densities within a year, and stomach fullness did not have a significant effect on increment width in our 2002 general linear model. While the stomach results suggest prey availability was not influencing salmon growth rate differences between habitats, it is possible salmon fed at different times or duration in the different habitats.

Other factors influencing the growth rate differences between habitats may include increased habitat area, fewer predators, and lower turbidity. Inundation of seasonal habitats (including floodplains, off-channel ponds, and tributaries) increases the overall wetted area available for fish and other aquatic organisms and may act to decrease competition for resources (Sommer et al. 2001). This same area effect may also act to dilute the number of predators and therefore lower the direct and indirect effects of predation. The presence of large predators may force smaller fishes into sub-optimal feeding areas (Schlosser 1987) and the growth rate differences may reflect different activity patterns in the different habitats. Large predators were absent from non-natal seasonal tributaries although we did find large-mouth bass in off-channel ponds (Limm pers. obs.).

Fine sediments can reduce growth rates in salmonids (Crouse et al. 1981, Suttle et al. *in press*). We observed higher turbidity in the main channel in both years although the differences were not significant. Berg and Northcote (1985) observed reduced feeding efficiency at 11 NTU, levels similar to main channel values in both 2001 and 2002. In addition to effects on growth, turbidity may influence selection of off-channel habitat during high flows (Scrivener et al. 1994).

Higher salmon growth rates in off-channel habitats may ultimately lead to higher survival. With juvenile salmon being both gape-limited predators and subject to gape-limited predation, faster growth can improve their ability as predators while decreasing their vulnerability to predation (Myrick and Cech 2000). In a study by Sommer et al. (2001) chinook salmon released into the Yolo Bypass floodplain grew faster and had higher adult return rates than those released into the main channel, suggesting that improved rearing conditions can increase survival rates. MacFarlane and Norton (2002) reported little growth and a decreasing condition factor for juvenile chinook salmon passing through the San Francisco estuary, and perhaps the larger size at entry decreases residence time and reduces vulnerability to predation.

Larger size has also been shown to improve competitive ability for optimal feeding positions (Fausch 1984). This benefit of off-channel rearing should extend to other native fish species as well. The shallow depth, warmer temperature, and more abundant prey in off-channel ponds and seasonal tributaries are conditions similar to those found in floodplain habitats. Floodplains are known to be associated with enhanced fish growth (Gutreuter et al. 2000, Sommer et al. 2001), higher secondary production (Gladden and Smock 1990), and a greater contribution of biomass to the river system (Benke 2001). While fish growth appears to be enhanced and overall prey densities were

over a magnitude higher in off-channel habitats, we do not know the total amount of biomass contributed to main channel communities from these alternative habitats.

Our results suggest river regulation and land-use policies may have significant impacts on juvenile salmon rearing and survival in California. Dams in the Central Valley watershed have dramatically altered the natural surface hydrology- and thermal regime of the Sacramento River. Winter precipitation and snowmelt runoff are responsible for the high flows and cooler water temperatures in the winter and spring, and almost all native species spawn during this time (Moyle 2002). High summer temperatures are believed to have limited main channel rearing by juvenile salmon to the winter and spring (Moyle 2002). The current altering of the natural hydrograph of the Sacramento River has decreased the connectivity of off-channel ponds in the spring and increased the flows (for Central Valley agriculture) in the summer. We observed stranding of juvenile salmon in both 2001 and 2002 as off-channel ponds were disconnected from the main channel. After losing connection, water temperatures likely increased to lethal limits ($\approx 24^{\circ}$, Rich 1987). If salmon life history strategies are adapted to utilizing off-channel habitats, the mortality costs associated with stranding may be significant. The increased flows and cooler water temperatures in the summer may provide nursery and rearing habitat for introduced species in connected off-channel ponds. Dettmers et al. (2001) found large-mouth bass almost exclusively in off-channel areas of the Illinois River, and we speculate that the current Sacramento River flow regime has improved rearing conditions for them.

Seasonal tributaries are threatened by land-use practices that alter the riparian vegetation and subsequent inputs of both carbon and surface/ground water. Due to their low overall discharge any water diversion or land-use practices that influence the

magnitude and duration of water run-off could impact their persistence and connectivity to the main channel (Bradford et al. 2001).

We conclude that off-channel habitats may play an important role in supporting juvenile salmon. Whether the higher growth rates observed in off-channel ponds and seasonal tributaries ultimately lead to higher survival rates is unknown. Future work assessing the impacts of off-channel rearing over the entire freshwater-ocean continuum is needed (Simenstad and Cordell 2000). Specifically, there is a need to quantify the number of salmon utilizing the various off-channel habitats and the mechanisms by which higher growth rates in the freshwater system influence survival through the estuary. We are encouraged by recent developments in otolith analysis. Their continuous growth and stable structure make them an ideal record of environment conditions in both space and time (Campana 2001). In the present study we use otolith microstructure to quantify higher salmon growth rates in off-channel habitats. Other uses of otolith microstructure include determining age (Campana and Neilson 1985), identifying distinct salmon life-history types (Zhang and Beamish 1995), and quantifying homing precision to natal incubation sites (Quinn et al. 1999). Recent developments in analyzing otolith chemistry have improved our ability to track fish populations (Campana et al. 1999) and may allow the reconstruction of a fish's life history (Weber et al. 2002). Using the practical knowledge of age, growth, and life history, we can improve predictive models and better understand how physical conditions influence salmon survival.

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Table 1. Results of physical measurements and salmon collections for 2001 and 2002.

	2001		2002		
	OP	MC	OP	MC	ST
No. of sites	1	2	3	3	2
Mean turbidity (NTU)	5.7±2.7	10.5±1.9	7.8±2.0	13.0±2.0	5.0±2.5
No. of salmon	30	60	90	90	60
Standard length (mm)	43.6±0.8	44.1±1.0	43.7±0.9	40.7±0.6	45.1±1.0
Mean salmon mass (g)	1.2±0.1	1.3±0.1	1.3±0.1	1.0±0.1	1.4±0.1
Spearman's Rank					
Correlation coefficient (r_s , p)	0.22, 0.38	-0.14, 0.47	0.23, 0.31	-0.17, 0.42	-0.24, 0.25

Table 2. Summary of general linear model to test the effects of habitat, prey density, stomach fullness, water temperature, and turbidity on increment width.

Dependent variable	Effects test	2001			2002		
		df	MS	F	df	MS	F
Increment width	Habitat	1	47.87	41.2 **	2	56.76	26.5 **
	Prey density	1	52.60	45.3 **	1	6.54	6.1 *
	Stomach fullness	1	9.75	8.4 *	1	0.88	0.82
	Water temperature	1	27.18	23.4 **	1	17.7	16.5 **
	Turbidity	1	11.14	9.6 *	1	0.88	0.8

Note: * indicates statistical significance at $\alpha = 0.05$, ** indicates statistical significance at $\alpha = 0.0001$.

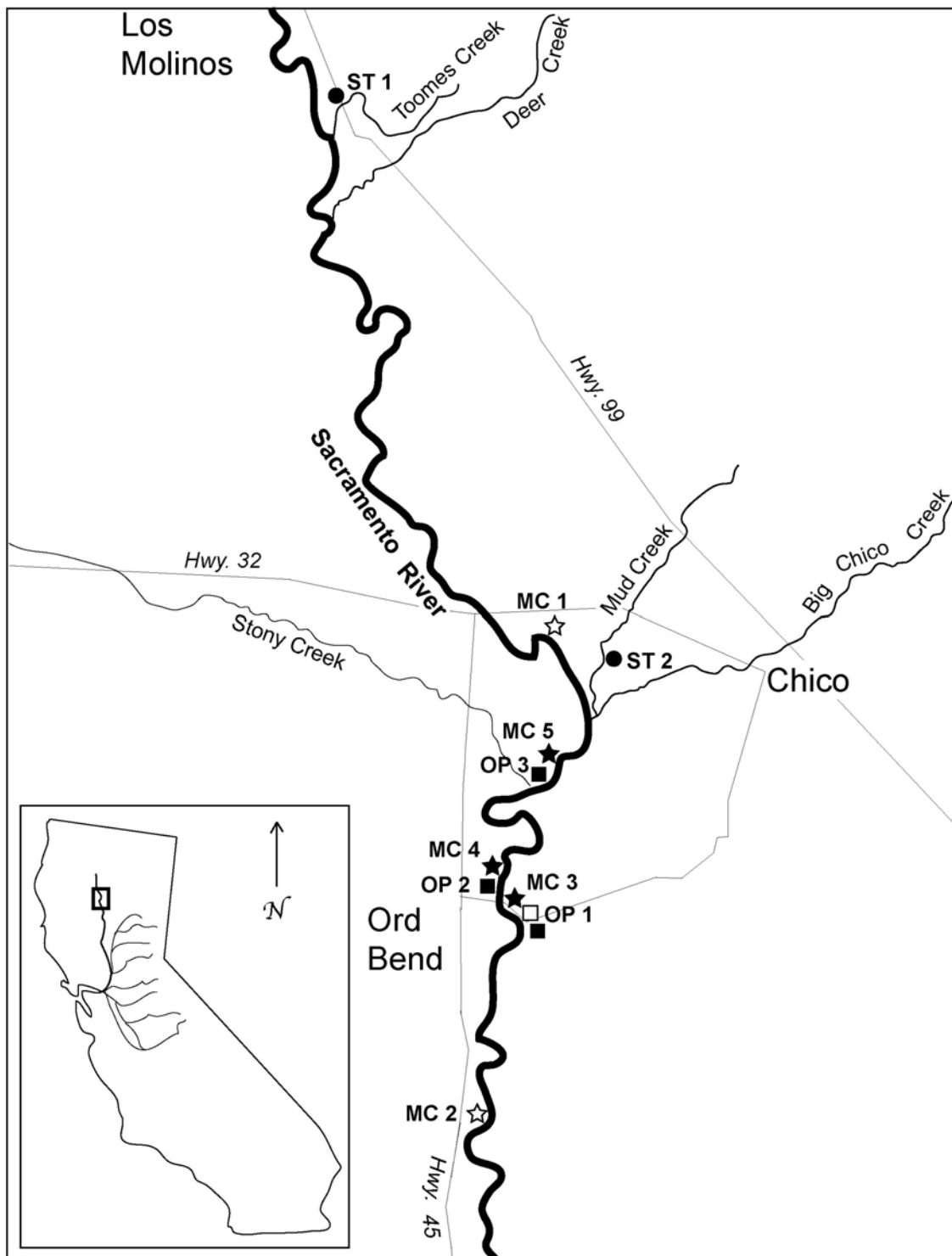


Fig. 1. Sampling sites for March-April 2001 (open symbols) and February-March 2002 (filled symbols) along the Sacramento River. Off-channel pond (squares), main channel (stars), and non-natal seasonal tributaries (circles) are shown for each year.

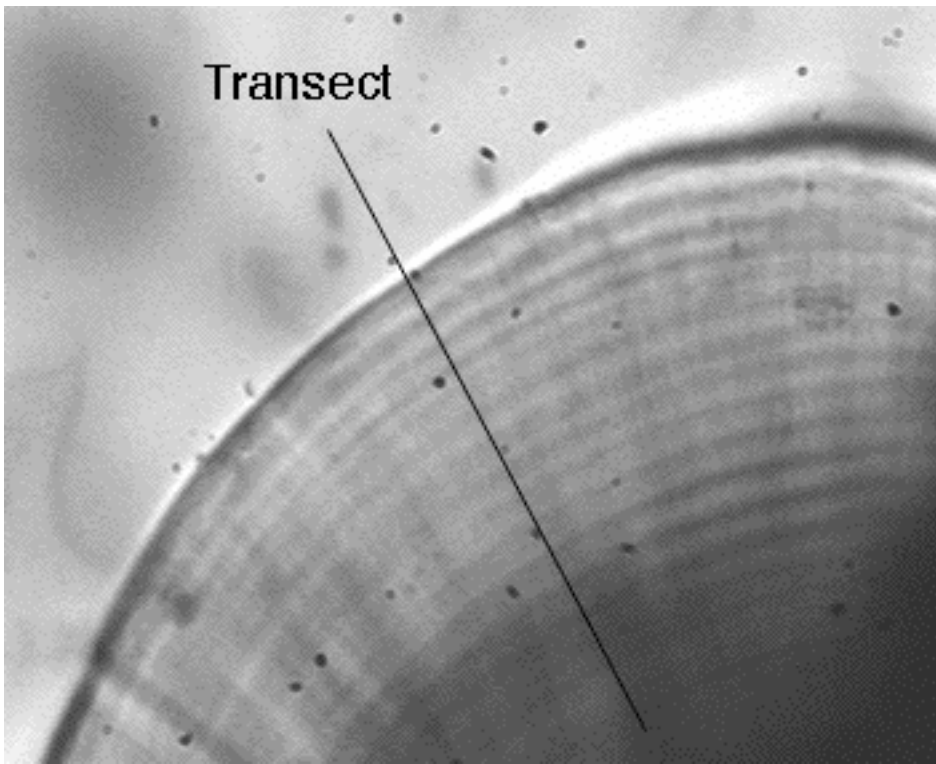


Fig. 2. Photograph of the anterior ventral region of a juvenile chinook salmon sagittae otolith taken at 400x. Measurements were made along the transect starting from the outermost recognizable increment.

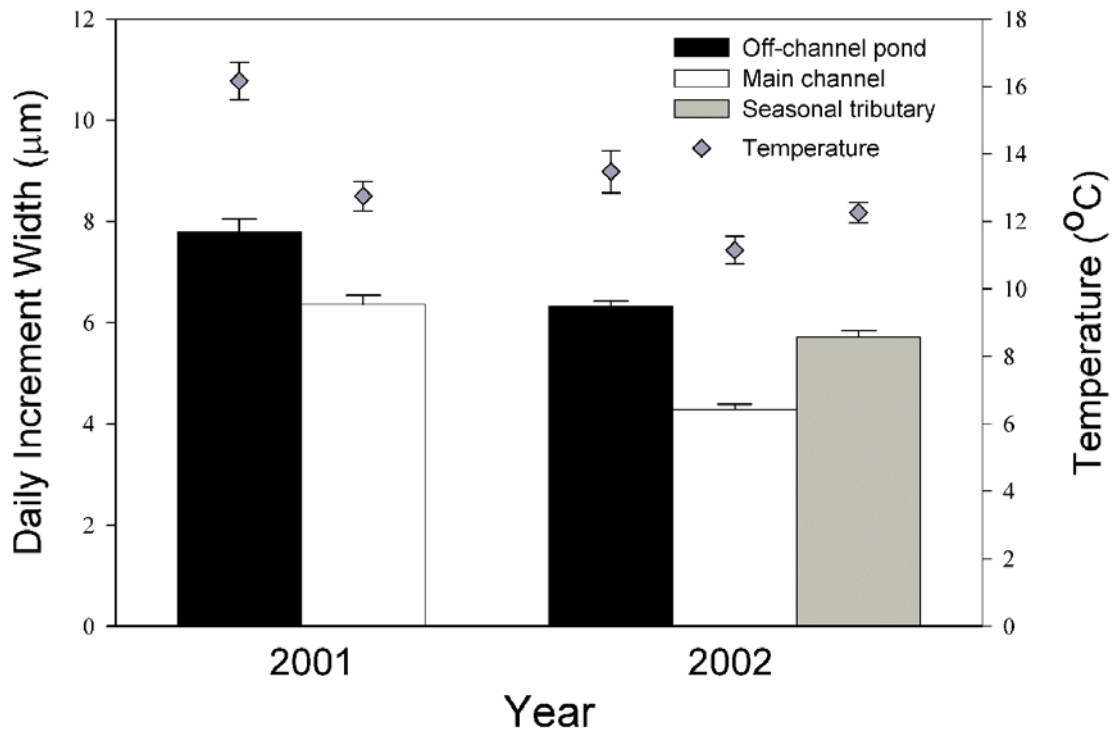


Fig. 3. Mean daily increment widths and temperature for different habitats in 2001 and 2002. Off-channel pond (solid bars), main channel (open bars), and non-natal seasonal tributaries (shaded bars) are shown for each year. The standard errors reported for increment widths and temperature are from nested ANOVA results.

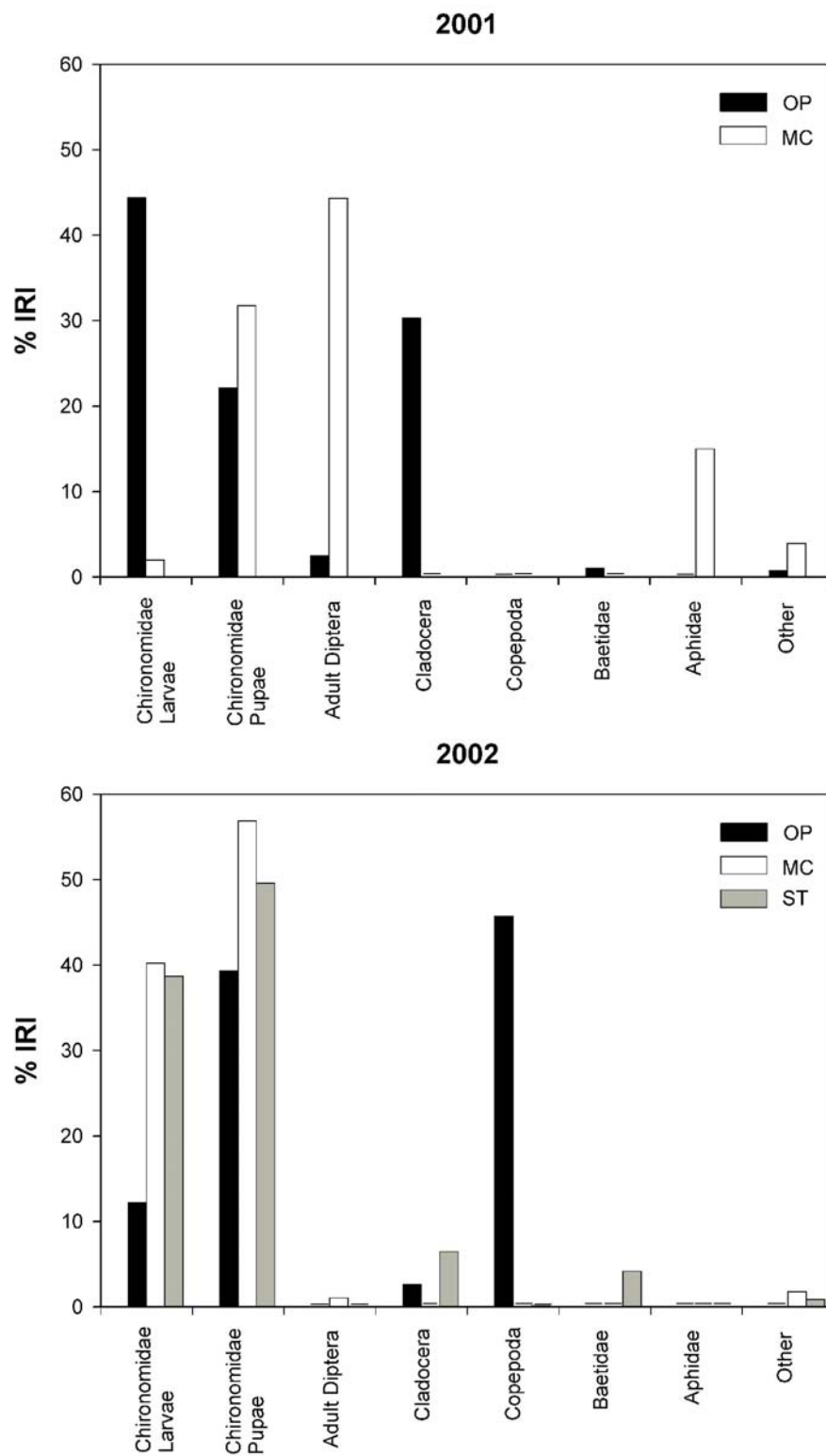


Fig. 4. Chinook salmon diet in off-channel ponds and the main channel during March-April of 2001 and in off-channel ponds, the main channel, and non-natal seasonal tributaries during February-March of 2002. The index of relative importance is defined in the text.

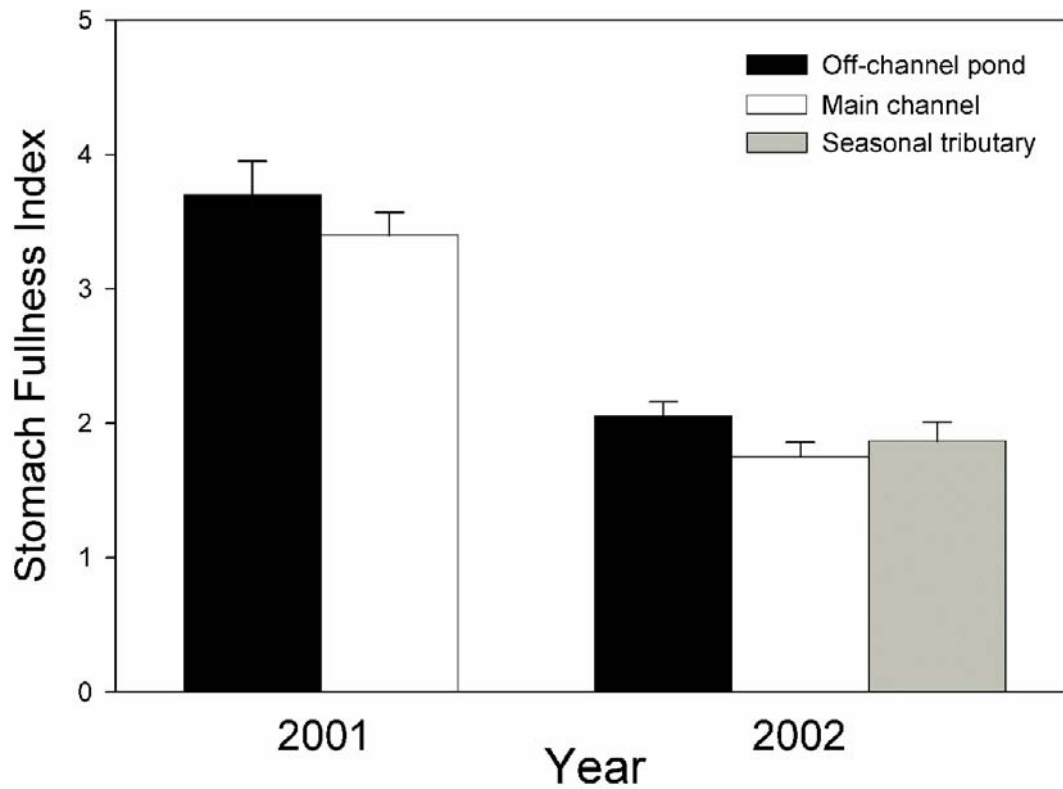


Fig. 5. Stomach Fullness Index for different habitats in 2001 and 2002. Off-channel pond (solid bars), main channel (open bars), and non-natal seasonal tributaries (shaded bars) are shown for each year. The standard errors reported are from nested ANOVA results.

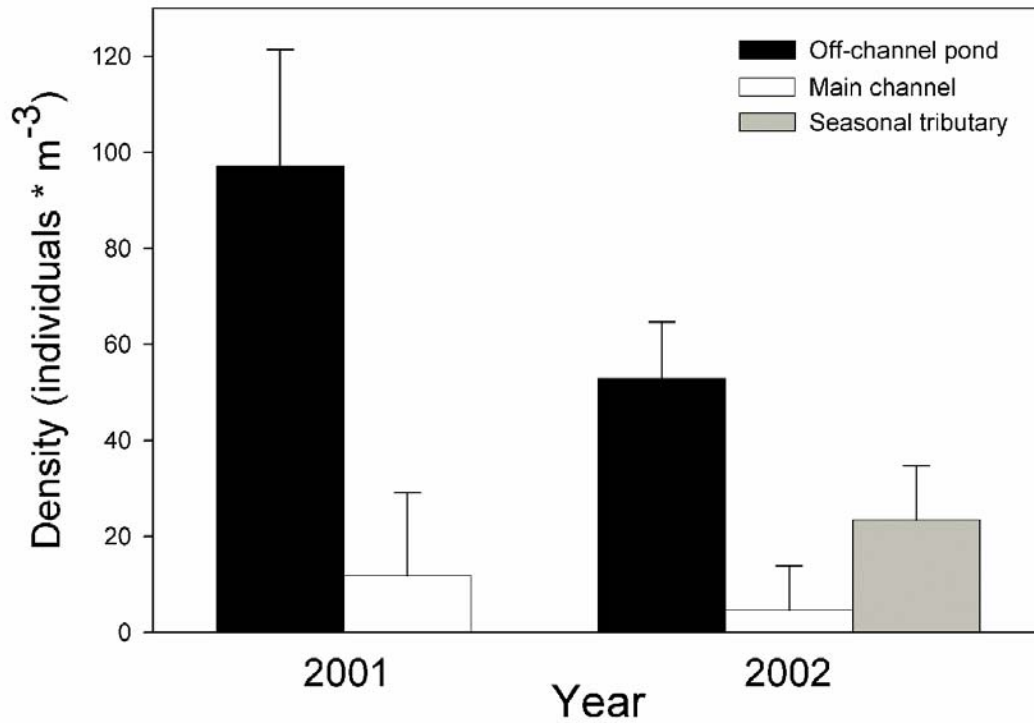


Fig. 6. Overall prey densities for different habitats in 2001 and 2002. Off-channel pond (solid bars), main channel (open bars), and non-natal seasonal tributaries (shaded bars) are shown for each year. The standard errors reported are from nested ANOVA results.

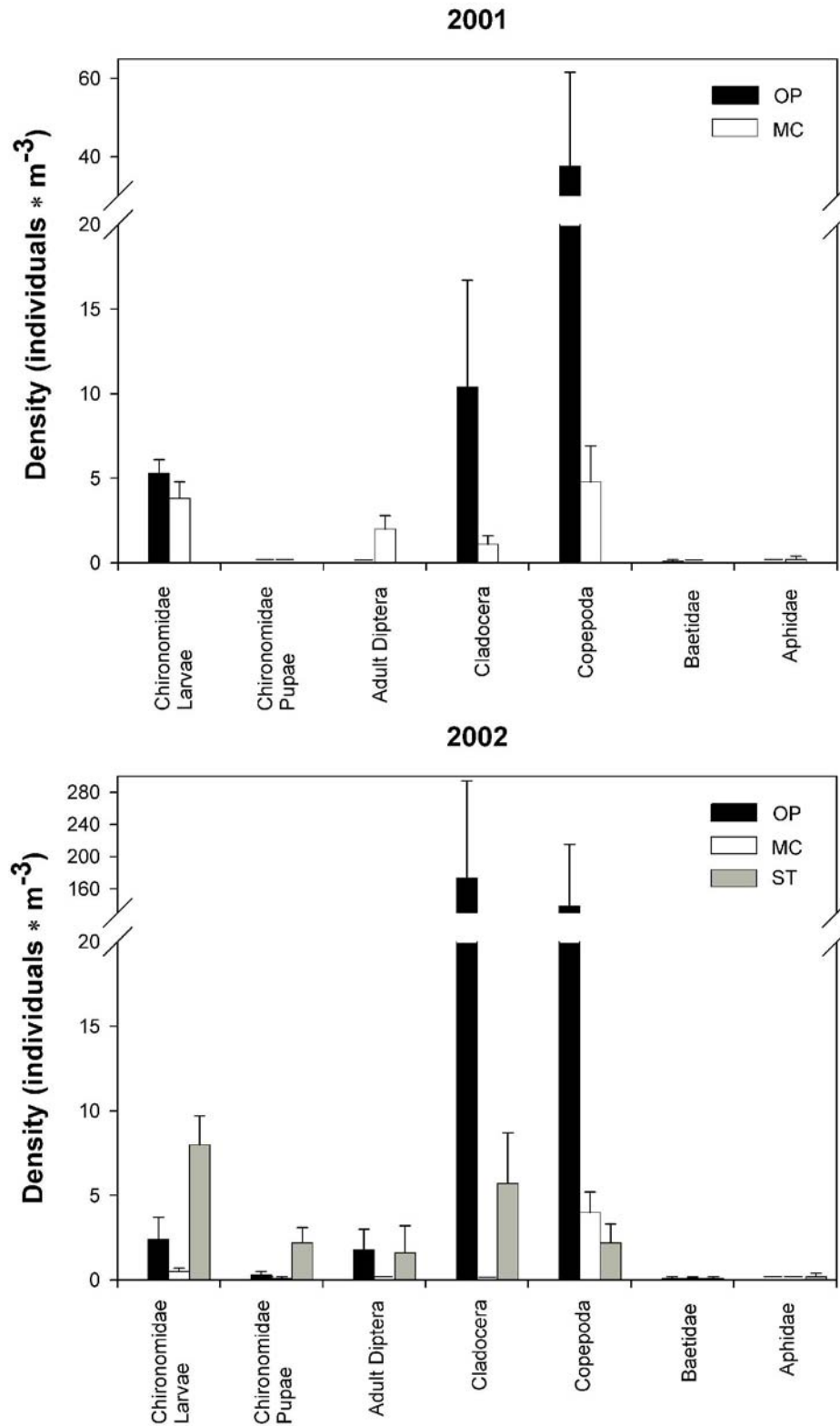


Fig. 7. Prey density by taxa for different habitats in 2001 and 2002. Off-channel pond (OP), main channel (MC), and non-natal seasonal tributaries (ST) are shown for each year. Organisms shown account for 96% IRI or greater of salmon diets. Means and standard errors are shown.